Introduction to Pattern Recognition

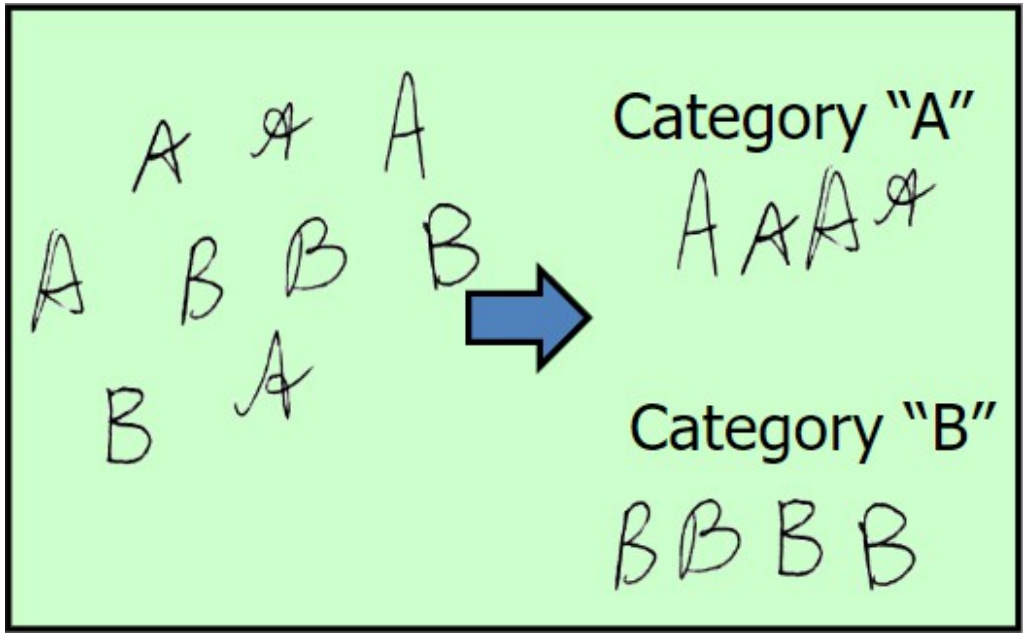
Chapter 01

Ref. Book: Pattern Classification - R. Duda et al.

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Pattern Recognition/Classification

Assign an object or an event (pattern) to one of several known categories (or classes).



What is Pattern

A set of instances that

- share some regularities and similarities
- is repeatable
- is observable, sometimes partially, using sensors
- may have noise and distortion

What is Pattern (cont.)

A set of instances that

- share some regularities and similarities
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What is Pattern (cont.)

- An object or event.
- Represented by a vector **x** of values corresponding to various features

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix}$$

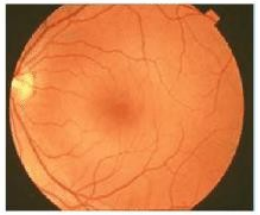
Some Biometric patterns



(a) Finger print



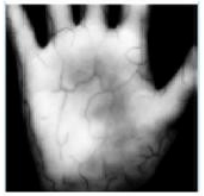
(b) Palm print



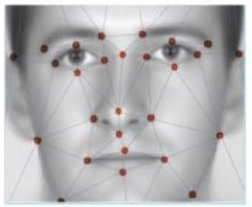
(c) Retinal vein



(d) Iris print



(e) Palm vein



(f) Face recognition

What is Class

A collection of "similar" objects



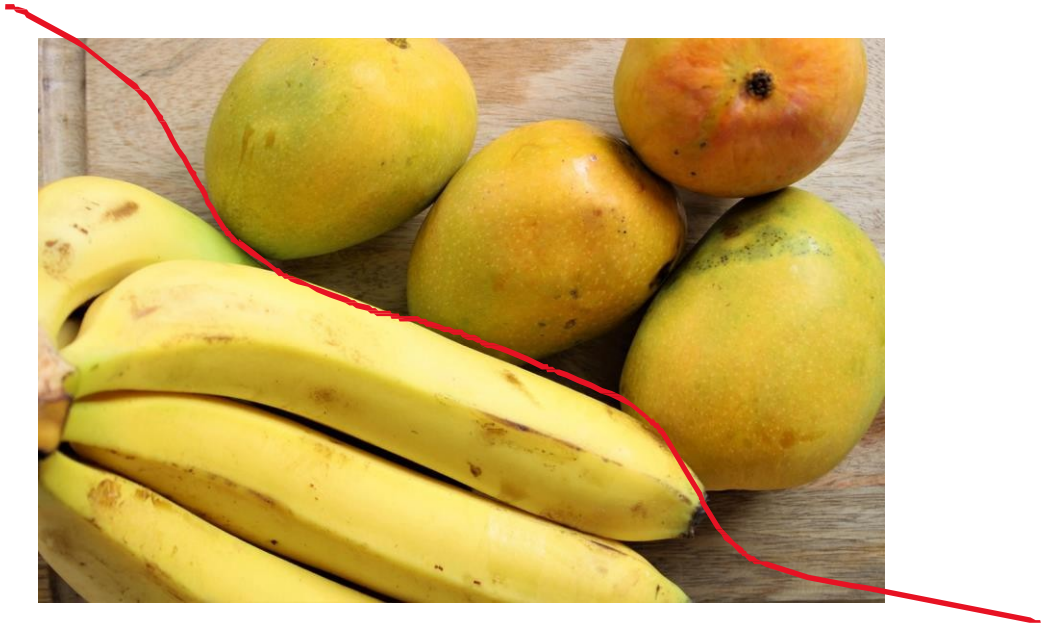
Class: Mangoes



Class: Bananas

Main Objective

- Separate data belonging to different classes.
- Assign new data to the correct class.



What is Pattern Recognition

- **Pattern recognition(PR)** is the scientific discipline that concerns the description and classification (recognition) of patterns (objects)
- The task: Assign unknown objects –> patterns –> into the correct class. This is known as classification.
- Typical application areas
  - Machine vision
  - Character recognition (OCR)
  - Computer aided diagnosis
  - Speech recognition
  - Face recognition
  - Biometrics
  - Image Data Base retrieval
  - Data mining
  - Bioinformatics

Representation of patterns

Features:

- These are measurable quantities obtained from the patterns
- The classification task is based on their respective values.

Feature vectors: A number of features

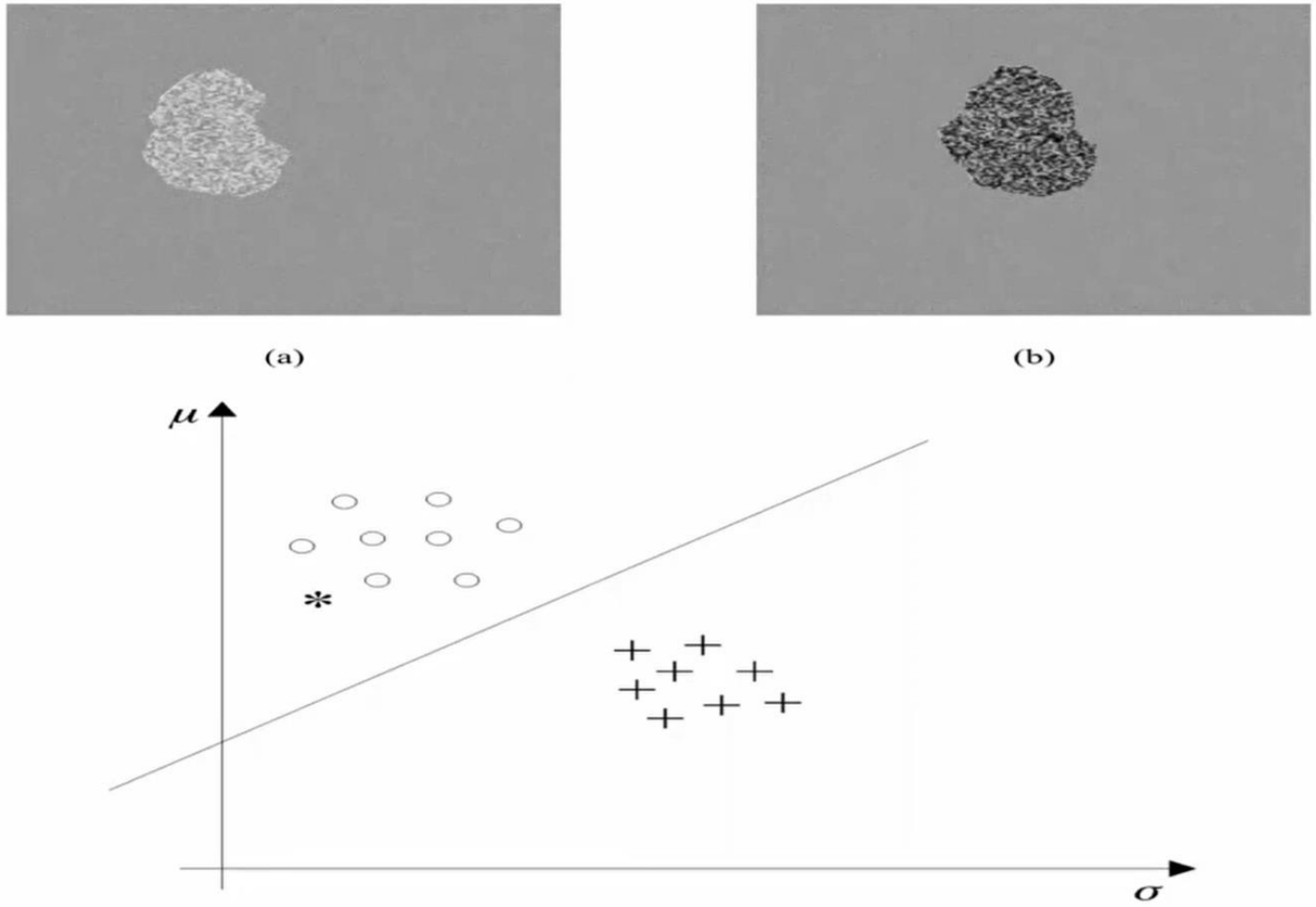
$$x_1, \ldots, x_l,$$

Constitute the feature vector

$$\underset{R}{\underline{x}} = [x_1, \ldots, x]^T \in {}^{l}$$

Feature vectors are treated as random vectors.

Example 01

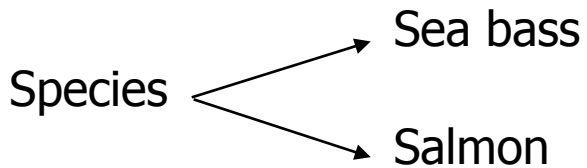


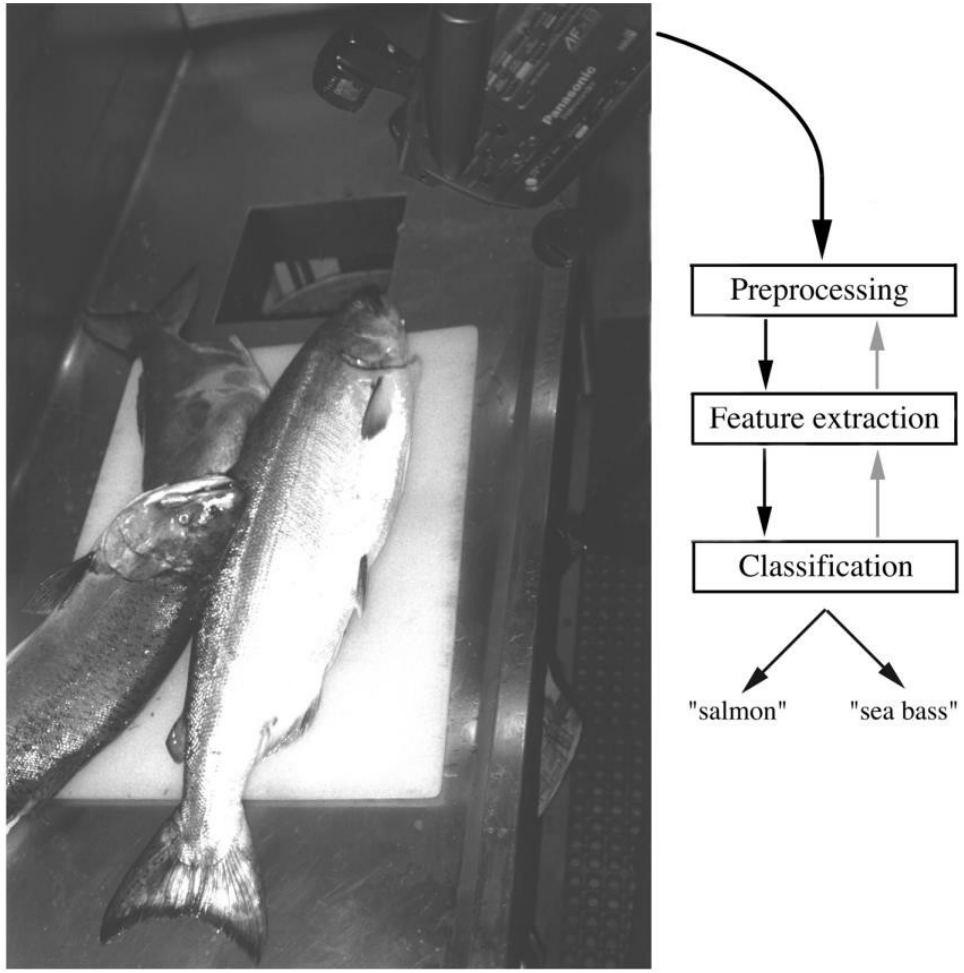
Issues in Pattern Recognition

- How are features generated?
- What is the best numbers of features?
- How are they used to design a classifier?
- How good is the classifier?

Example 02

"Sorting incoming Fish on a conveyor according to species using optical sensing"





Problem Analysis

Set up a camera and take some sample images to extract features

- Length
- Lightness
- Width
- Number and shape of fins
- Position of the mouth etc….

This is the set of all suggested features to explore for use in our classifier!

Main Steps

Preprocessing:

- Isolate fishes from one another and also removes background

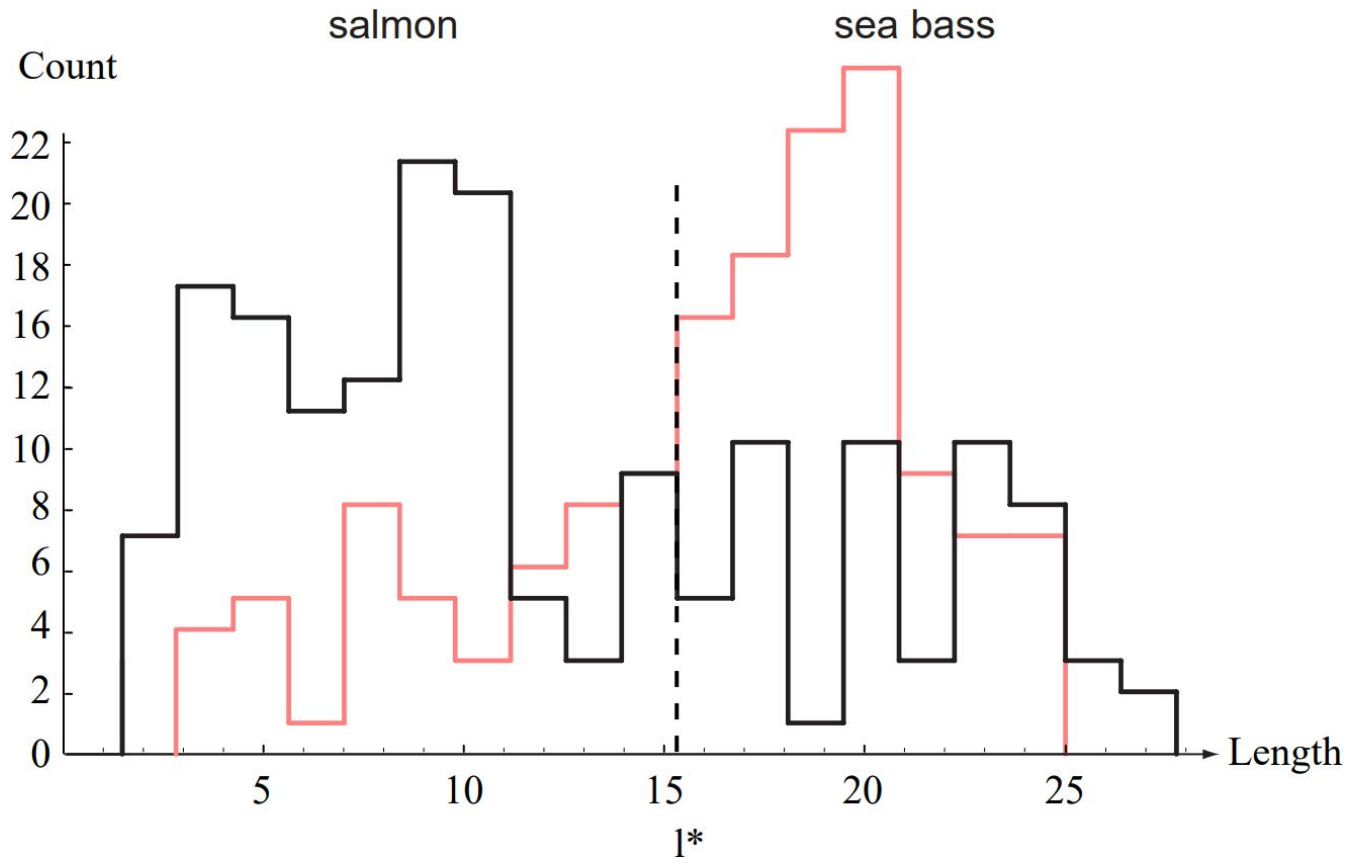
Feature extraction:

- Send isolated fish image to feature extractor
- It reduces data too

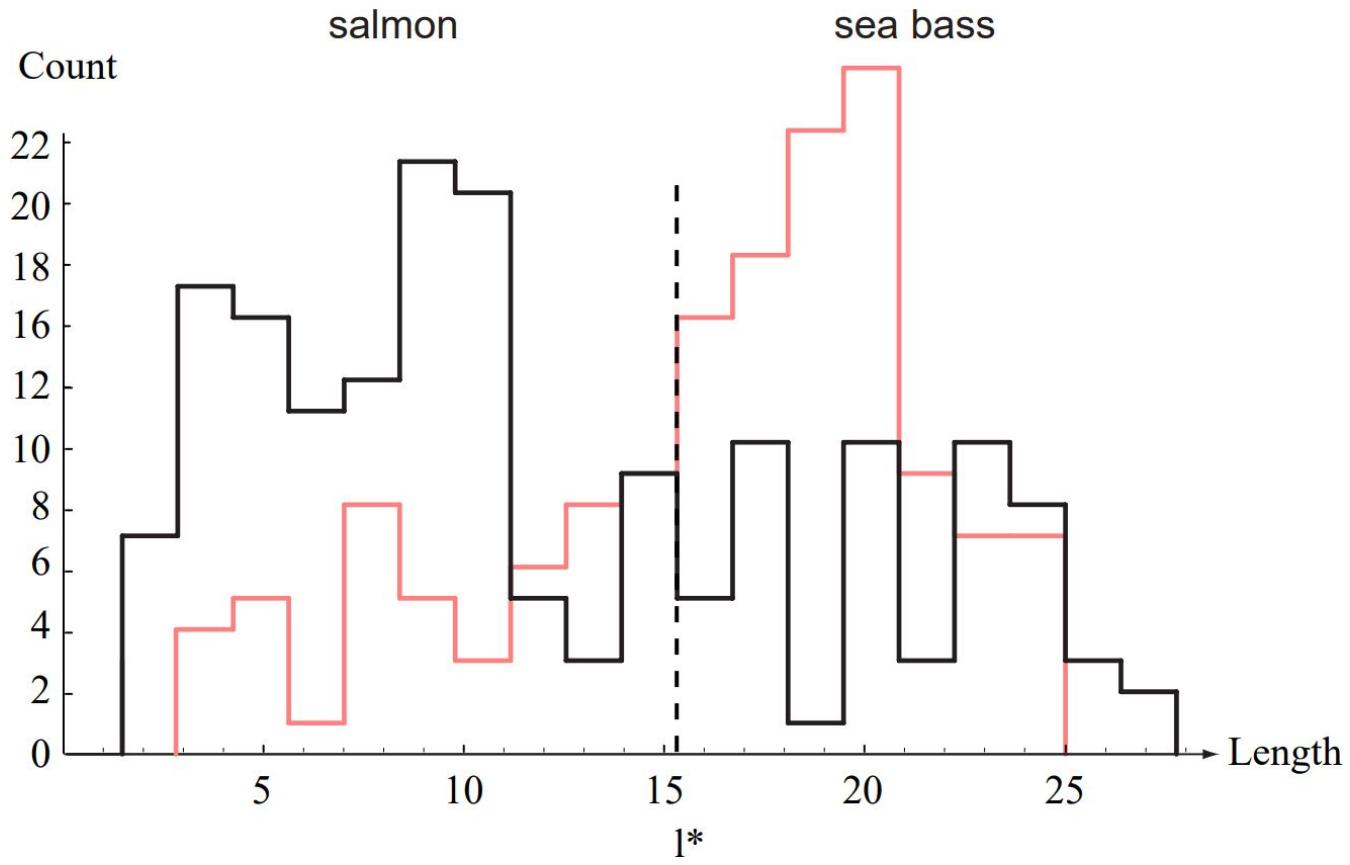
Classification:

- Pass the feature to a classifier

Length as a discriminator

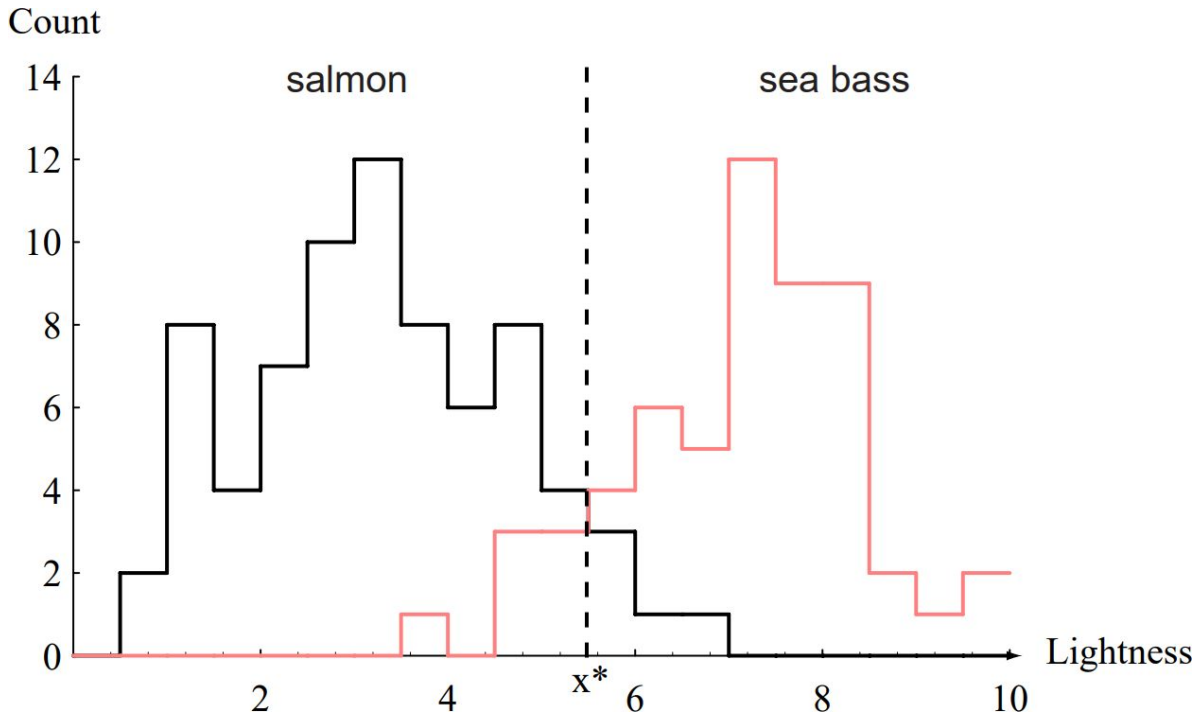


Length as a discriminator

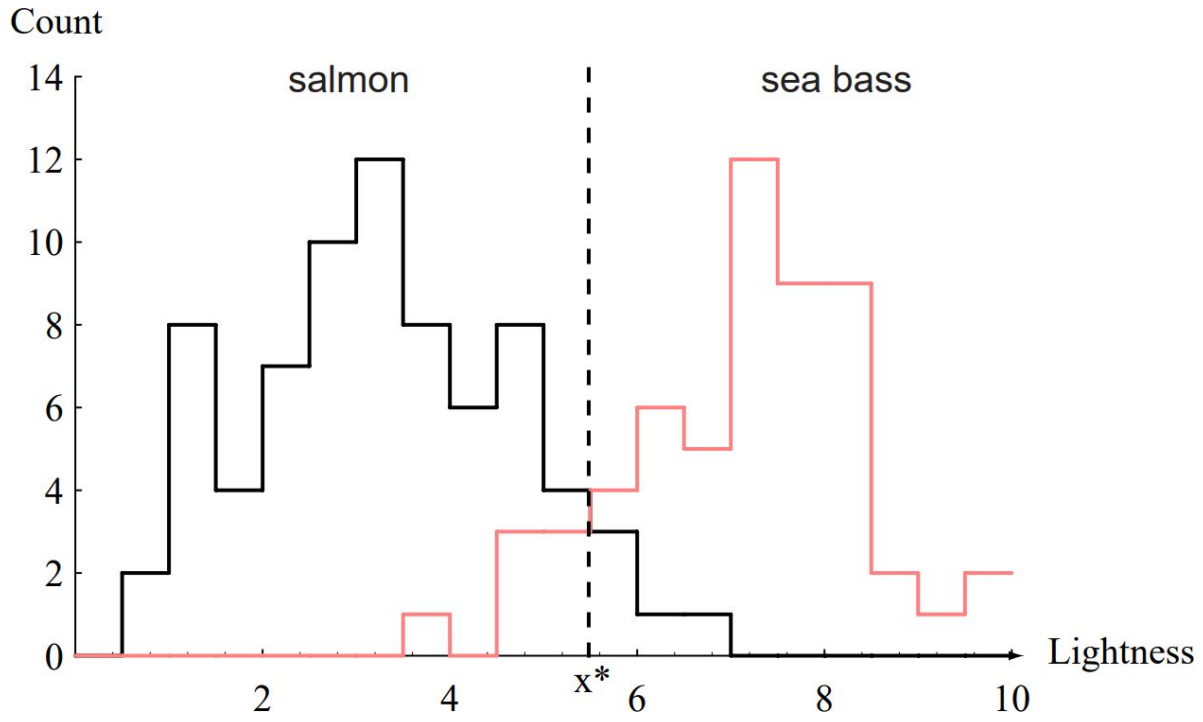


Conclusion: Length is a poor feature alone

Lightness as a discriminator



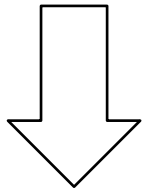
Lightness as a discriminator



- **Lightness** is a better feature than length because it reduces the **misclassification error**.
- Can we **combine** features in such a way that we improve performance? (Hint: **correlation**)

Decision Boundary and Cost Relationship

Move our decision boundary toward **smaller** values of lightness in order to minimize the cost (reduce the number of sea bass that are classified salmon!)



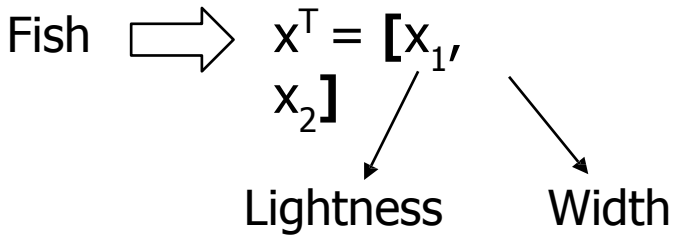
Task of **decision theory**

Multiple features

To improve recognition accuracy, we might need to use more than one features.

- Single features might not yield the best performance.
- Using combinations of features might yield better performance.

Adopt the lightness and add the width of the fish



Multiple features

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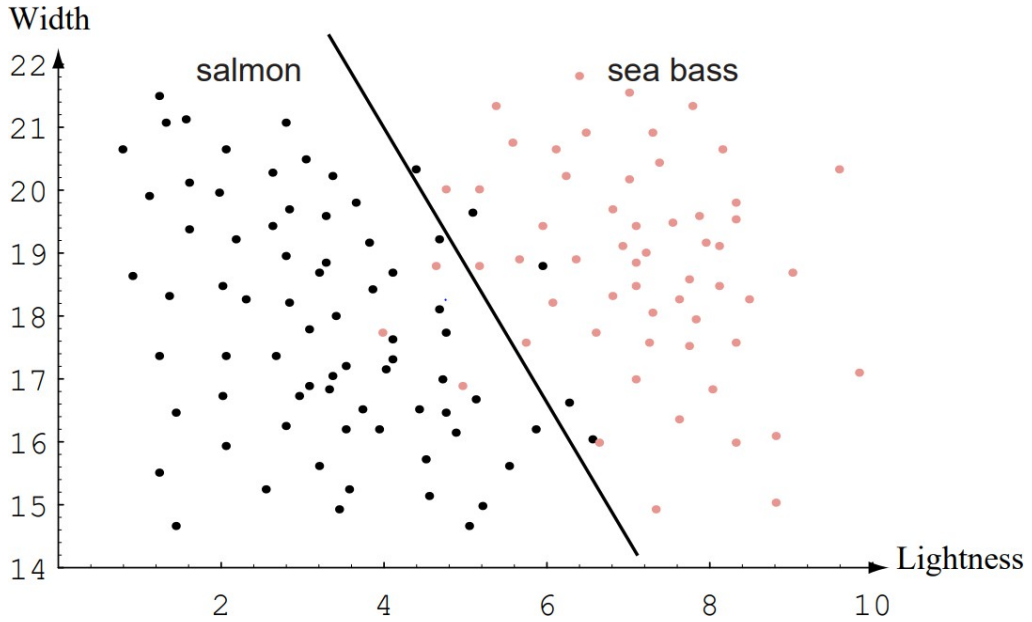
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Adopt the lightness and add the width of the fish

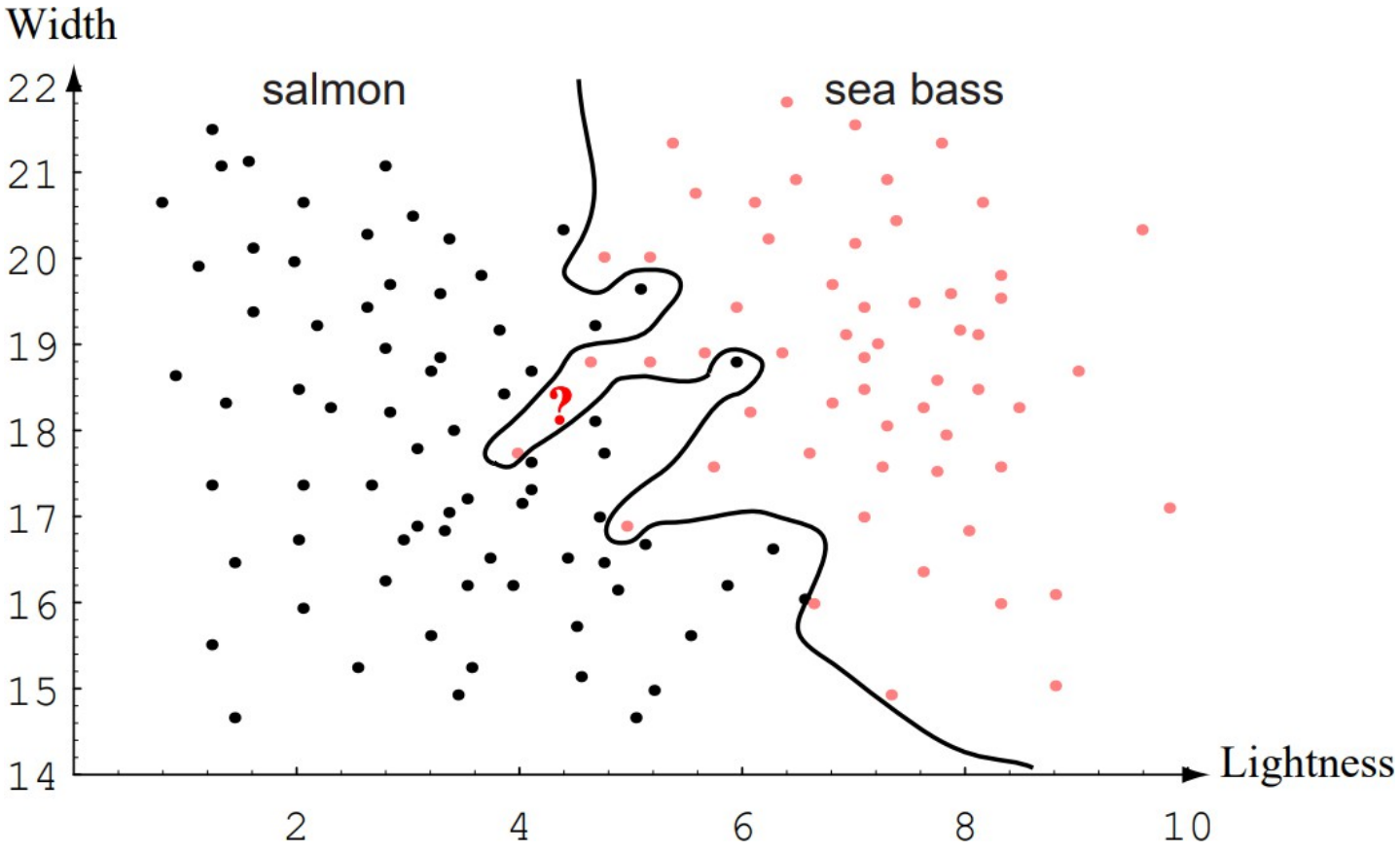
Fish ⇨ $x^T = [x_1, x_2]$

x_1: Lightness

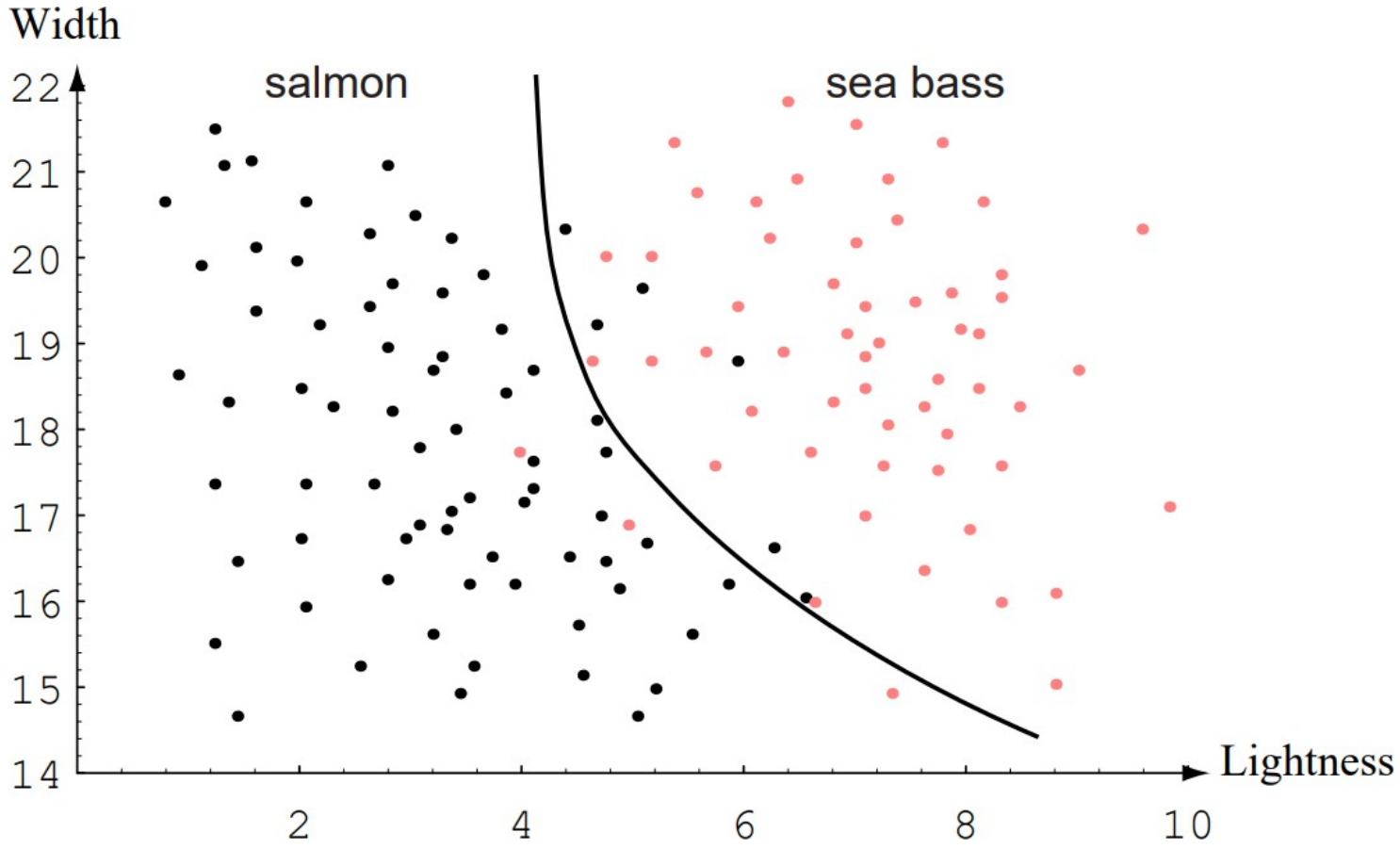
x_2: Width



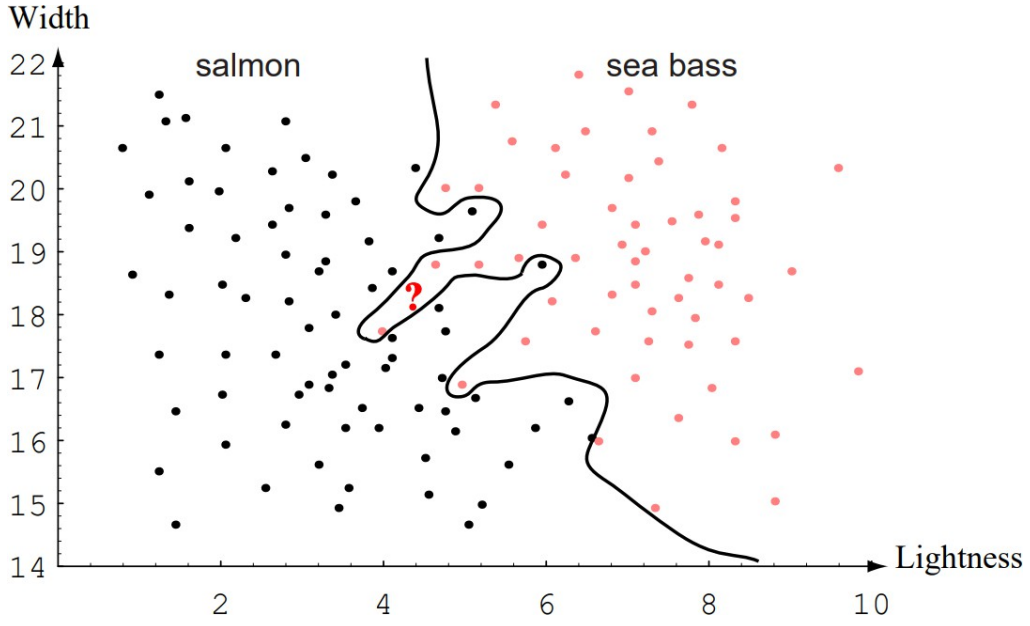
Complex decision boundary



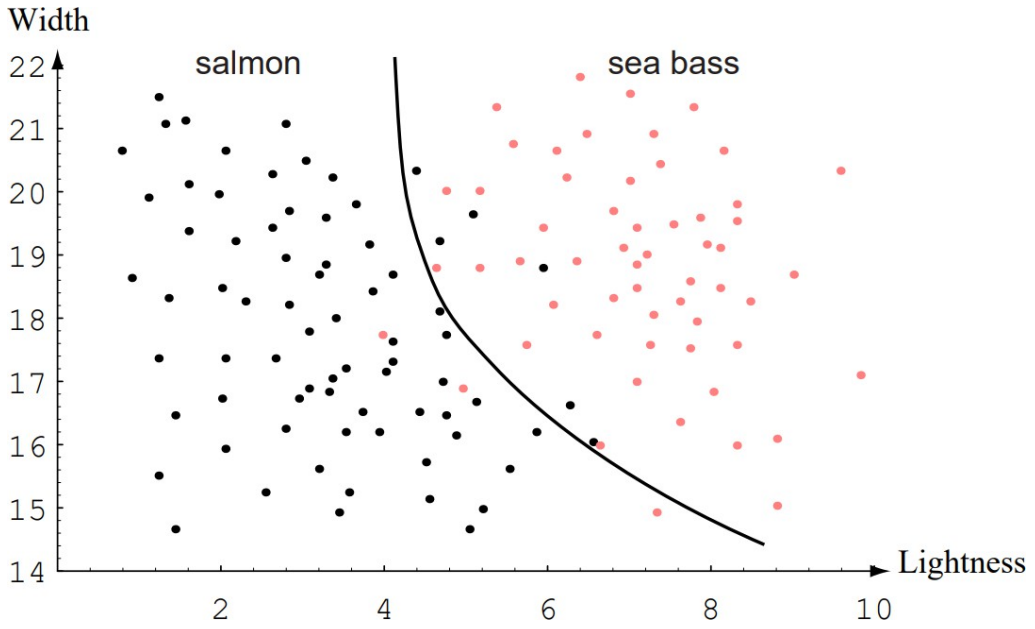
Compromise between training and testing



Complex model vs Simple model

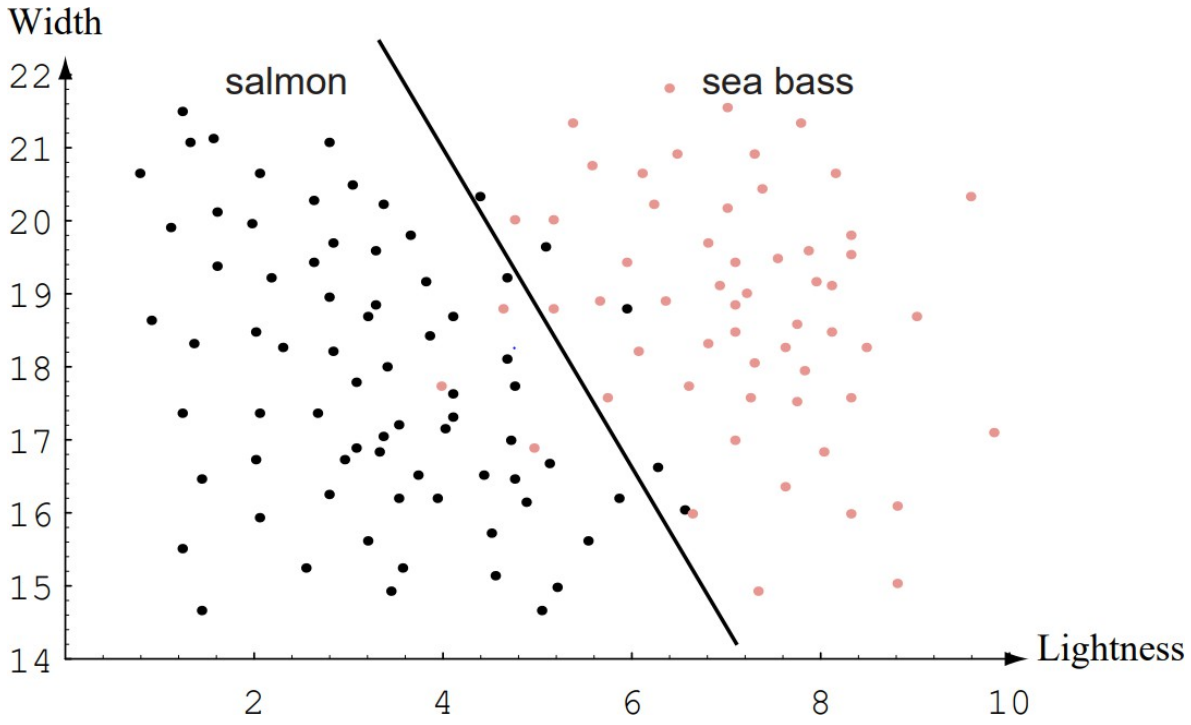


Overfitting



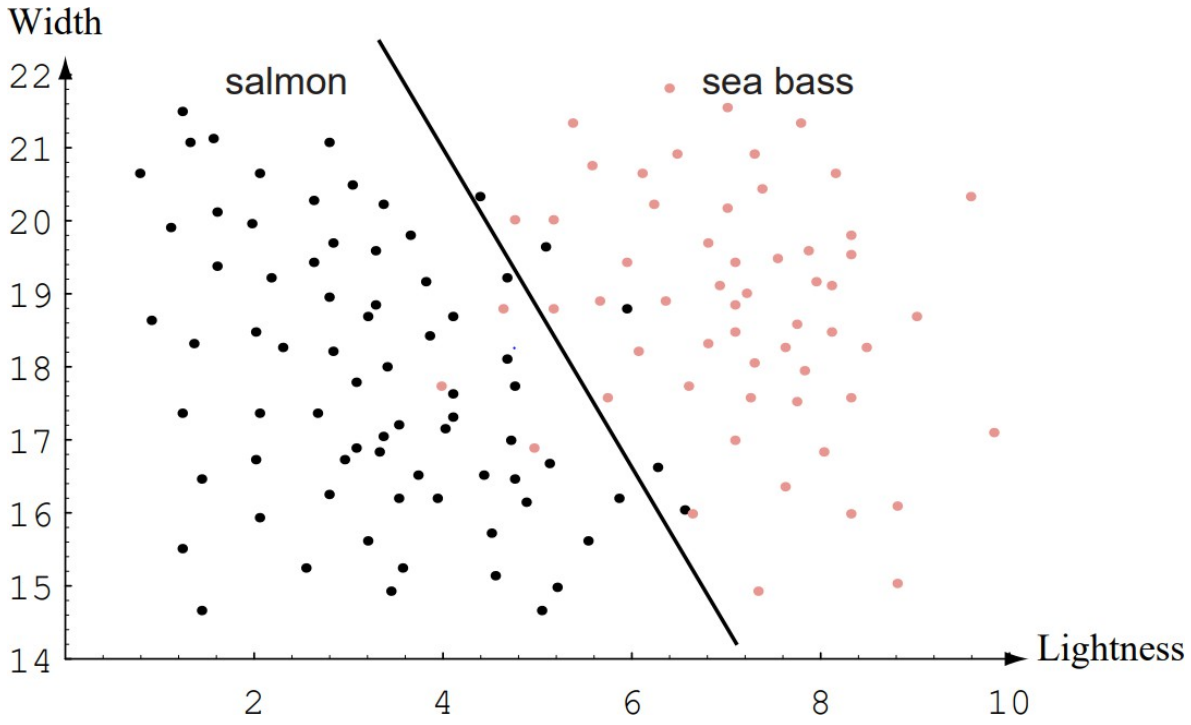
Generalization

Complex model vs Simple model (cont.)



What is this?

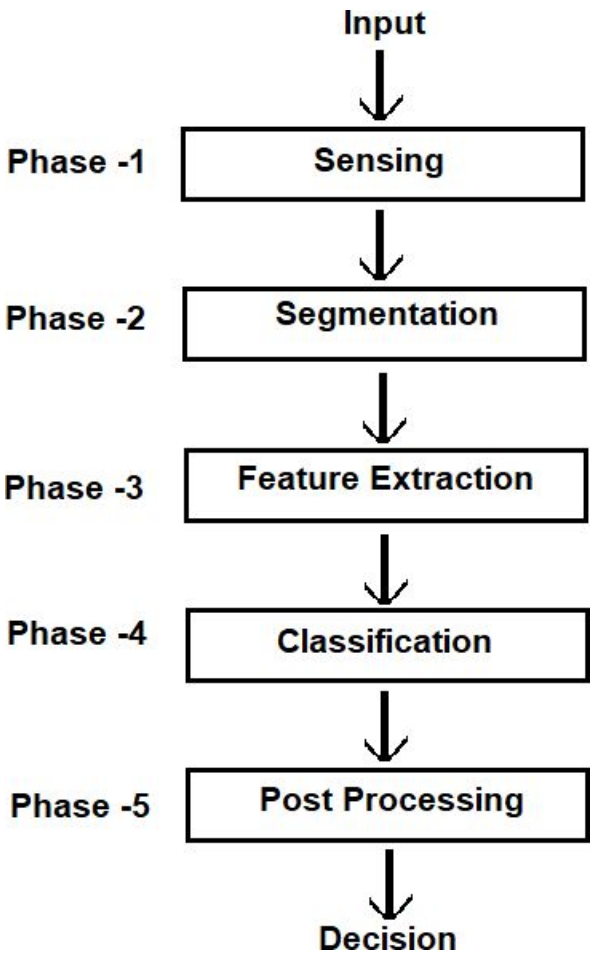
Complex model vs Simple model (cont.)



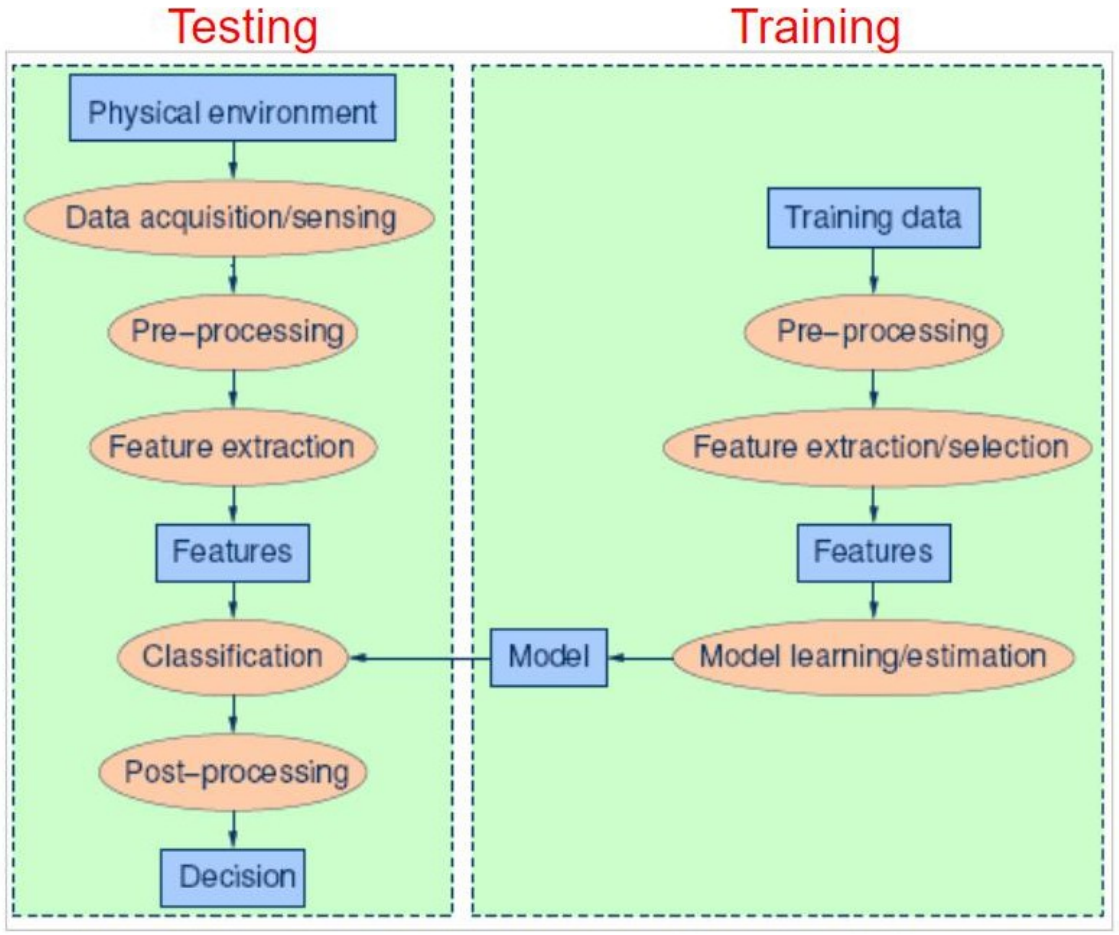
What is this?

Underfitting

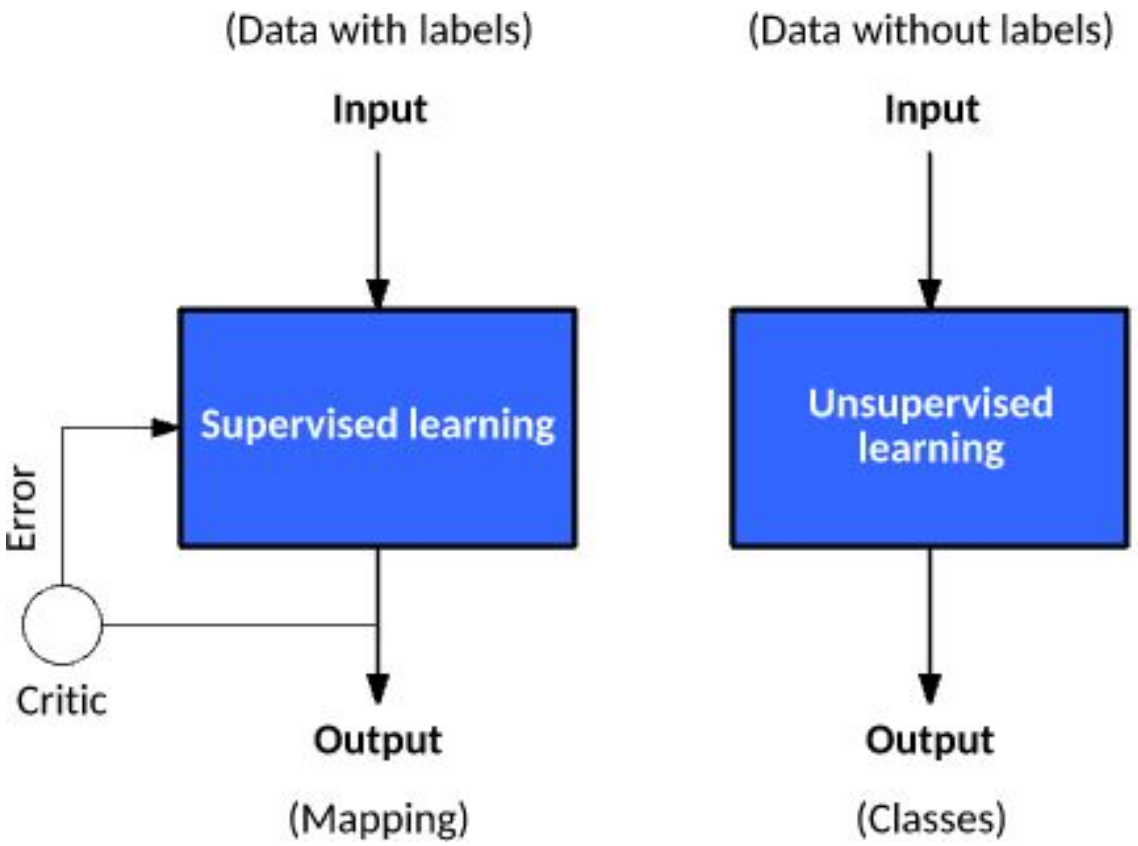
Design Cycle (Simplified)



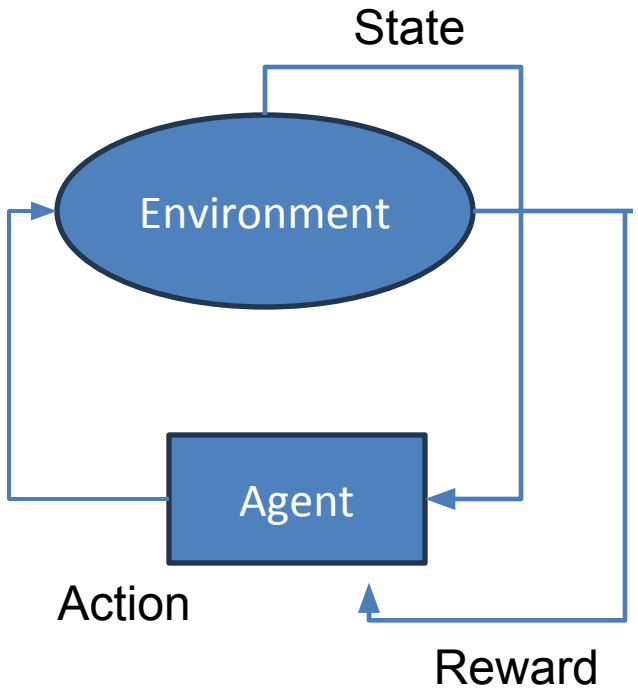
Design Cycle (Advanced)



Learning

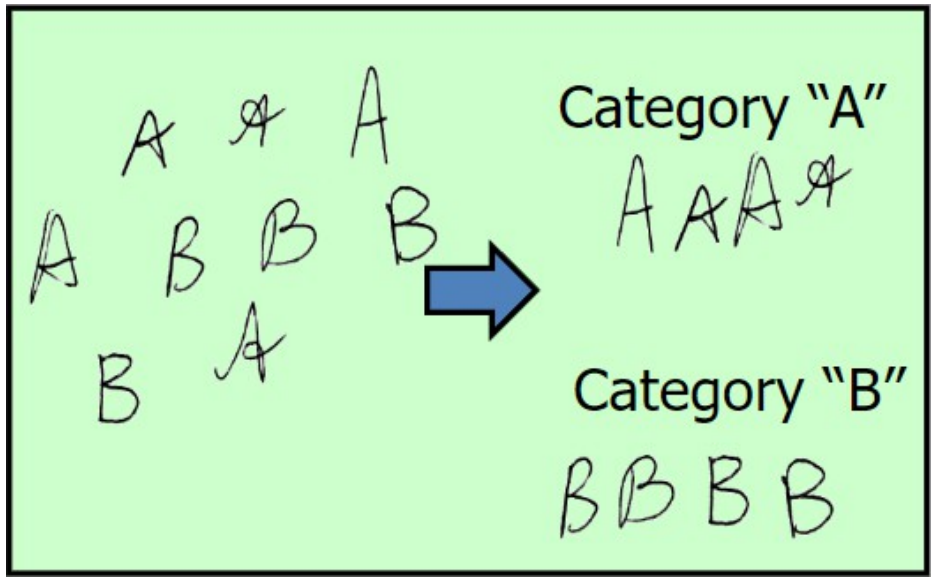


Reinforcement Learning

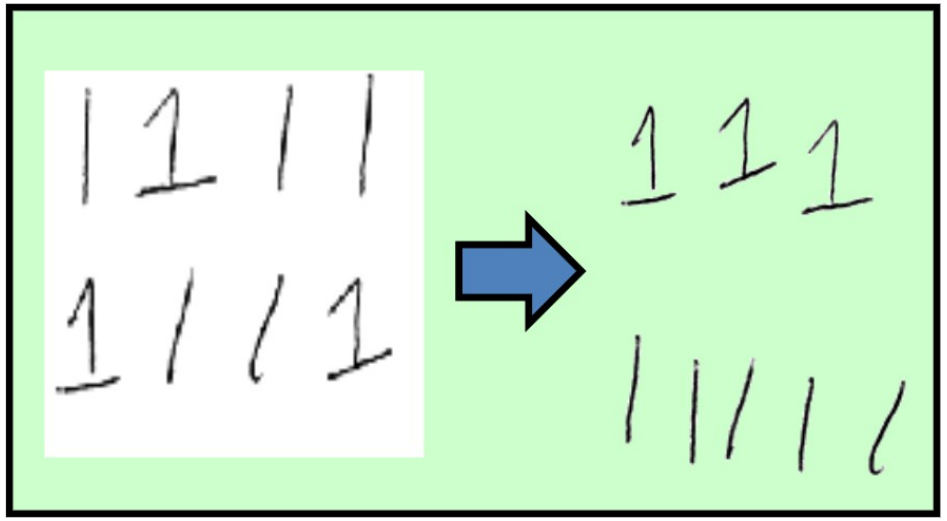


Supervised vs Unsupervised

Classification (known categories)

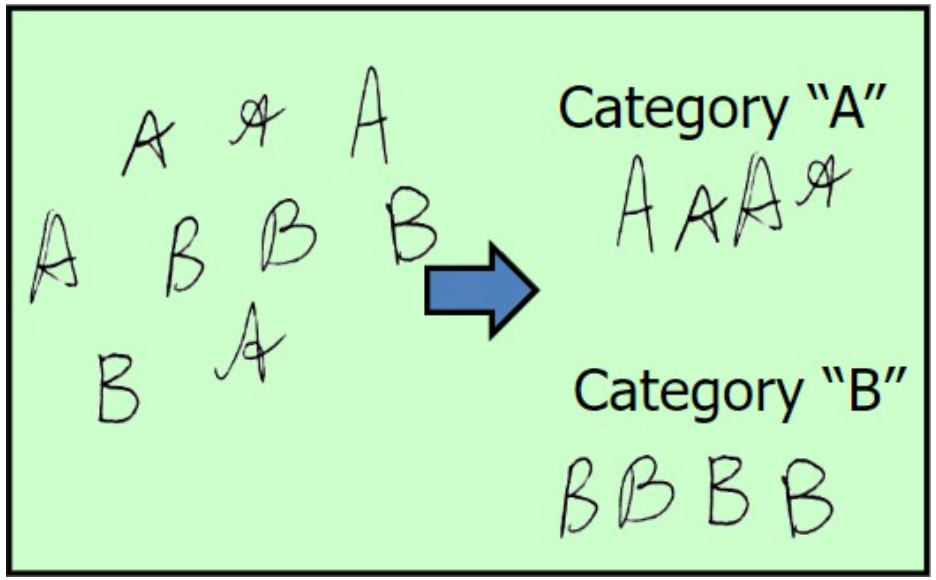


Clustering (unknown categories)

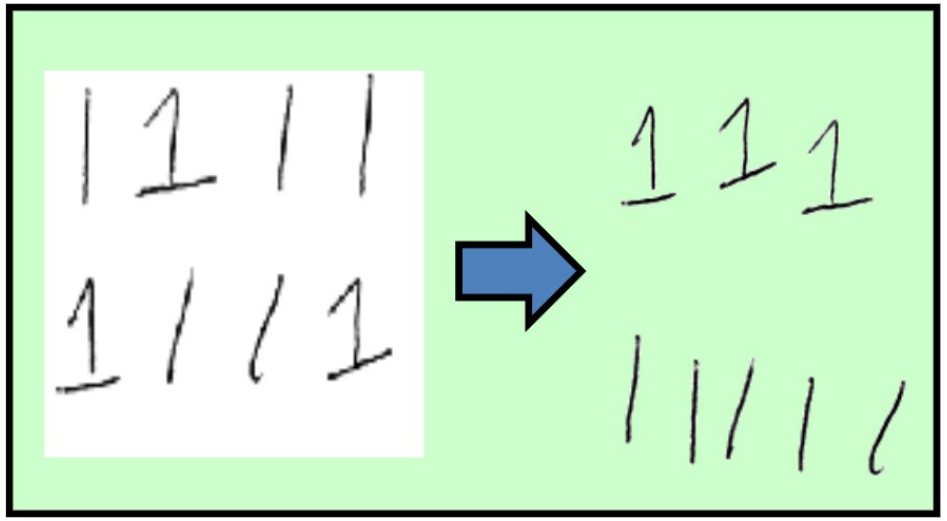


Supervised vs Unsupervised

Classification (known categories)
Supervised



Clustering (unknown categories)
Unsupervised



The End